

SIE Exam Prep Notes

Gabriel

September 4, 2026

Contents

1	Knowledge of Capital Markets	3
1.1	Market Participants	3
1.1.1	Broker-Dealers	3
1.1.2	Broker-Dealer Departments	3
1.1.3	Market Makers	4
1.1.4	Investment Advisers	4
1.1.5	Types of Investors	5
1.2	Market Structure	5
1.2.1	Primary Market	6
1.2.2	Secondary Market	6
1.3	Other Venues	6
1.3.1	Third Market	7
1.3.2	Fourth Market	7
1.3.3	Electronic Communication Networks (ECNs)	7
1.3.4	Dark Pools	7
1.4	Clearinghouses and the DTCC	7
1.4.1	Clearing Firms	7
1.4.2	Introducing Firm Accounts	8
1.4.3	Prime Brokerage	8
1.4.4	Clearing Options Contracts	8
1.5	Other Market Entities	8
2	Understanding Products and Their Risks	10
2.1	Individual Securities—Equities	10
2.2	Individual Securities—Debt	10
2.3	Derivatives	10
2.4	Packaged Investments	10

2.5	Other Investment Vehicles	10
2.6	Types of Risk	10
3	Understanding Trading, Customer Accounts, and Prohibited Activities	11
4	Overview of Regulatory Framework	12

1 Knowledge of Capital Markets

1.1 Market Participants

One of the primary purposes of the securities industry is to match investors, who have money, with issuers that need money.

Definition 1.1 (Issuer). An issuer is a legal entity that sells securities in order to finance its operations. Examples of issuers are government agencies, governments themselves, corporations, and banks. Issuers primarily issue either debt securities (bonds) or equity securities (stocks).

Debt securities are publicly traded loans, referred to as bonds, notes or debt instruments. A bond represents the amount of indebtedness (principal) that the issuer owes to the investor. The investor (purchaser) is considered a *creditor* and has lent their money to the issuer for the specified period (till maturity). The bond issuer is expected to pay the principal balance at the end of maturity, as well as possible interest payments. If any payment is missed then the bond is in *default*.

Equity securities represent ownership interest of a corporation, and are issued by corporations to raise capital. If the company is profitable then those with ownership interest may be entitled to profits (dividends). This security does not typically have a maturity date and dividend payment is voluntary for the issuer.

Financial firms connect issuers and investors, and are broadly categorized into either *broker-dealers* or *investment advisers*.

1.1.1 Broker-Dealers

A broker engages in the business of *agency* transactions in securities for the account of others. In an agency transaction, the broker acts strictly as an intermediary connecting buyers and sellers without taking ownership of the securities into inventory, thereby avoiding principal risk and earning compensation through commissions rather than markups.

A dealer is a person that engages in the business of buying or selling securities for its own account. Dealers engage in *principal* transactions, where securities are bought directly from clients and held in inventory, or sold to clients from inventory.

Since firms can continuously execute trades as either a broker or a dealer, it is convenient to refer to the whole lot as broker-dealers.

1.1.2 Broker-Dealer Departments

Brokerage firms (broker-dealers) are divided into many distinct departments. The most common of which are:

- Investment Banking (IB): IB is an area that works directly with issuers to arrange and structure their securities offerings. Investment bankers are often referred to as *underwriters* of securities, and may also assist companies with mergers or acquisitions (M&A) or restructuring due to bankruptcy.

- Research: Analysts study the market and individual issuers to make recommendations (typically buy, sell, or hold).
- Sales: Market individual stocks or bonds (or packaged products such as mutual funds) to retail investors and institutions. Historically referred to as stock or bond brokers, but more recently as *registered representatives (RRs)* or *investment advisor representatives (IARs)*.
- Trading: Handles the execution of trades for the firm's clients and the firm's own proprietary account. Trades can occur in electronic marketplaces (e.g. Nasdaq) or hybrid marketplaces (e.g., NYSE).
- Operations: Ensures all paperwork, funds, and securities transfers associated with trades or processing is handled efficiently and in compliance with industry standards and regulations. Functions may include generating customer statements, confirmations, tax records, and assisting in transfer of funds or securities. Also responsible for ensuring that client and firm assets are organized and safeguarded.

1.1.3 Market Makers

Market makers post quotes (two-sided) for buyings and/or selling securities at specific prices. Often, market makers are *required* to make regular bids (buy requests) and offers (sell offerings) and to meet specific capital requirements. For example, where the firm is willing to buy 1,000 shares at 17.05 and/or sell 2,000 shares at 17.08:

Bid	Ask	Size
17.05	17.08	1,000 × 2,000
Price at which the firm will buy	Price at which the firm will sell	

Traders are under no obligation to enter quotes into the marketplace, instead they execute trades against the quotes of market making firms.

1.1.4 Investment Advisers

Investment advisers (IAs) differ from broker-dealers in how they are compensated. Broker-dealers earn commissions for executing transactions, whereas IAs charge fees for providing service to their clients, regardless of whether any trades occurred in the client's account. Investment advisers also often assist investors in building a balanced portfolio through a process that's referred to as asset allocation.

IAs can manage hundreds of millions of dollars or have small practices working with individual investors. The size of the assets under management (AUM) determines who the IA must register with:

- < \$100 million: registered in the states they conduct business
- \$100-\$110 million: register with SEC or states
- > \$110 million: must register with SEC

A municipal advisors provide advice to or on behalf of a municipal entity (state, county, city, etc.). Their clients are typically the issuer rather than the investor. They provide advice on the structure, timing, and terms of a municipal finance offering. This term includes many different specific roles, so long as they are involved in municipal advisory activities.

1.1.5 Types of Investors

There are two primary types of investors, *retail investors* and *accredited investors*.

Retail investors have limited assets and income and buy stocks and bonds directly from broker-dealers. This can be for themselves or for a spouse or friend (joint account). Other forms of retail ownership include retirement accounts (e.g., IRAs) or custodial accounts for children. Much of securities regulation is aimed at protecting retail investors. Often, retail investors are considered those that do not meet the definition of an institutional investor.

Accredited investors are those who are categorized by the SEC as able to assume greater risk (whether that be due to greater assets or sophistication of trading strategy/trading infrastructure), and includes:

- Financial institutions, large tax-exempt pension plans
- Directors, executive officers, or general partners of the issuer
- Individuals with certain professional certifications
- Individuals with net worth $> \$1$ million (excluding primary residence) *or* having gross income greater than \$200,000 individually or \$300,000 joint with spouse, for the last two years and with expectation of continuance

Institutional Investors

Institutional investors are typically large entities that pool their money to purchase securities. These include include banks, insurance companies, pension plans, endowments, and hedge funds. The Financial Industry Regulatory Authority (FINRA) also includes individuals or entities with assets $\geq \$50$ million as institutional accounts.

There are also *qualified institutional buyers (QIBS)*, which can only be entities (not individual humans) that pass three requirements:

1. Is one of: insurance company, registered investment companies, registered investment advisers, small business development companies, private and public pension plans, certain bank trust funds, corporations, partnerships, business trusts, and certain non-profit organizations.
2. Is purchasing for its own account or the account of another QIB
3. Owns and invests at least \$100 million of securities from issuers *not* associated with the buyer

By specifying which category each investor belongs to (institutional and non-institutional), regulators can create rules which apply to each separately.

1.2 Market Structure

The securities market is structured in such a way that securities are issued in one market (the primary market) and traded in another (the secondary market).

1.2.1 Primary Market

As stated earlier, the primary market is the place where shares are *initially* issued. Companies will often pay a fee to the IB department of a brokerage for underwriting services, which involves marketing the shares to the investors.

1.2.2 Secondary Market

After the primary distribution of the issuer's shares, the investors will likely want to sell again. This is done on the secondary market, where the issuer is no longer involved and the funds pass between investors.

Traditionally, stock markets were either physical trading venues (NYSE) or over-the-counter (OTC) marketplaces. These centralized exchanges function as auctions. The auctioneer controlling trading is the designated market maker (DMM). Now, most modern exchanges are hybrid, where trading occurs on a physical floor but also electronically.

Equity securities that meet the standards for trading on a national exchange are referred to as *listed securities* and considered to be National Market System (NMS) securities.

The most recognized dealer-to-dealer network is the Nasdaq, the National Association of Securities Dealers Automated Quotation System.

Dealer-to-Dealer Market

When stocks don't qualify to be listed on exchanges, they instead trade OTC and are referred to as *OTC equities* or unlisted securities. The OTC Market Group provides quotes and organizes them into four market tiers based on quality, timeliness, and completeness of issuing company disclosure information:

1. OTCQX Best Market: highest tier. Companies meet strict financial and regulatory requirements, with annual and quarterly reports with the SEC (or a bank regulator or international regulator).
2. OTCQB Venture Market: for entrepreneurial and development-stage companies (domestic and international). Minimum bid requirement of \$0.01. Financial standards are less stringent than OTCQX.
3. OTCID Basic Market: new (launched July 2025). For companies that provide ongoing financial disclosures but cannot meet requirements of QX/QB.
4. Pink Limited Market: minimal to no issuer involvement. Highest risk category in the OTC market structure, and individual broker-dealers may place additional trading restrictions on Pink Limited securities.

Corporate, municipal, and U.S. government bonds don't have an organized exchange and are considered *non-equities*. Most bonds are traded in the OTC market through dealer-to-dealer networks.

1.3 Other Venues

Alternatives to traditional exchanges have been created:

1.3.1 Third Market

The third market refers to exchange-listed securities traded over-the-counter or otherwise away from traditional exchanges. This brings together investors and also supports after-hours trading.

1.3.2 Fourth Market

This refers to direct institution-to-institution trading that doesn't involve public markets or exchanges.

Since these two markets are typically transactions between dealer-brokers and large institutions (third) or between large institutions (fourth), they have little impact on typical stock trading for an average investor.

1.3.3 Electronic Communication Networks (ECNs)

ECNs are market centers supporting both quoting and trading of exchange-listed securities, providing an electronic system for matching buyers and sellers anonymously. ECNs charge a subscriber fee and act only in an agency capacity (broker).

1.3.4 Dark Pools

A system that provides liquidity for large institutional investors and high-frequency traders but does not disseminate quotes. Since details of the quotes are concealed from the public, dark pools allow the specific investors to buy and sell large blocks of stock with little market impact and with low transaction costs.

1.4 Clearinghouses and the DTCC

Once trading terms are agreed, the actual funds and security still need to be transferred between the parties. This is usually handled by a third part called a clearinghouse.

The Depository Trust & Clearing Corporation is a securities depository and a national clearinghouse for the settlement of securities (e.g., bonds, money-market instruments, derivatives). The DTCC automates and centralizes the clearing and settlement of trades among its members. Most financial institutions are members of the DTCC system. The DTCC is a non-profit and industry-owned. It is regulated by the SEC and the depository is a member of the U.S. Federal Reserve System.

The National Securities Clearing Corporation (NSCC) and the Fixed Income Clearing Corporation (FICC) are both subsidiaries of the DTCC. The NSCC clears equity trades for both U.S. and foreign issuers, while the FICC clears bond trades.

1.4.1 Clearing Firms

Clearing a trade has significant complexity that requires significant investment in hardware, software, and personnel. For this reason smaller broker-dealers often hand off all or part of trade processing to larger broker-dealers.

The firms who don't clear their own trades are called *introducing firms*. Their clients consider them to be their broker-dealer, but the clients funds and securities are physically held at the larger firm (called the *clearing firm*).

The clearing firms sit one step below the DTCC (also called *full-service firms*). These perform order execution, clearing, and settlement.

1.4.2 Introducing Firm Accounts

The basis for which a clearing firm provides services to an introducing firm can be either *fully disclosed* or *omnibus*.

Fully-disclosed accounts share all customer details with the clearing firm, who makes a separate account for each customer and handles all the relevant paperwork. The paperwork is identifiable as coming from the clearing firm but has details to connect it to the introducing firm.

Omnibus accounts keep the individual customer details hidden, so that the clearing firm handles just a single account for the introducing firm. In this arrangement, the introducing firm handles the paperwork.

1.4.3 Prime Brokerage

Prime brokerage service allows the clearing firm to hold all the positions for the customer in one centralized location, instead of requiring them to create and manage different accounts with each broker-dealer. Prior to this, the customer would need to take each statement and combining the information to understand their overall position. This consolidates bookkeeping.

1.4.4 Clearing Options Contracts

Options are derivative trading products that track the value of an individual stock. They can be purchased at an options exchange or OTC. Options contracts purchased on an exchange are called listed options.

Listed options trade on The Chicago Board Options Exchange (CBOE), The Boston Options Exchange (BOX), The NYSE Arca, The Nasdaq PHLX (formerly the Philadelphia Stock Exchange), and The International Securities Exchange (ISE).

Exchanged-traded options are standardized, meaning the terms are set and uniform. The DTCC doesn't clear these, instead these are cleared by the Options Clearing Corporation (OCC). Broker-dealers deal with the OCC directly, and must settle options contracts with the OCC within one business day.

1.5 Other Market Entities

There are several other entities that keep the market running smoothly:

- Custodians: banks or other financial institutions that hold the issuers' securities for safekeeping for the customers. Assets can be held physically, but securities are often held digitally (*book entry*).
- Registrars & Transfer Agents: issuers of publically traded securities need a company to keep track of the owners of their stocks and bonds. Firms providing the recordkeeping often act as both the registrar and transfer agent.

- The registrar maintains the ownership register of the issuer for each of its securities. Also ensures that the number of issued shares doesn't exceed the number of authorized shares.
- The transfer agent handles issuance and cancelation of certificates to reflect change in ownership, acting as a paying agent for bond interest payments or stock dividends, as a proxy for voting and mailing, and handling lost or stolen certificates
- Securities Trustees: some types of investments need a trustee to hold security interests that are created on trust for the benefit of various creditors. They may also make sure that issuers abide by promises (covenants) in a formalized agreement called a *trust indenture*. They also represent investors in the event of default and/or bankruptcy.

2 Understanding Products and Their Risks

2.1 Individual Securities—Equities

Definition 2.1 (Security). A security is an intangible financial asset that may be bought, sold, or gifted between persons. It may be represented by a paper certificate or held in an electronic record.

2.2 Individual Securities—Debt

2.3 Derivatives

2.4 Packaged Investments

2.5 Other Investment Vehicles

2.6 Types of Risk

3 Understanding Trading, Customer Accounts, and Prohibited Activities

4 Overview of Regulatory Framework